

Day 8: Chrome/Edge MV3 Extension Architecture & In-Gmail Shield

Key Milestone: Browser Extension Manifest V3 & In-Page Injection

- Sprint Focus: Execution of core deliverables for Day 8 spanning AI/NLP Intent Classification, Email Header Forensics, IOC Intelligence, Relationship Graphing, Case Workflow, and Full-Stack Frontend.

Key Deliverables Completed: MV3 Extension Package, Gmail DOM Ingestion Script, Extension Popup Controller.

Individual Team Member Task Logs & Technical Contributions

- Muniswami.Y — AI / NLP Lead

Domain Ownership: Explainable AI, Intent & Phishing Detection, Social Engineering signals | Key Modules: NLP/AI Engine, Explainability, Content Risk scoring

Day 8 Execution: Optimized lightweight heuristic model to run client-side in extension background worker for zero-latency offline threat scoring.
- Jeevan reddy.G — Header & Forensics Lead

Domain Ownership: Email header analysis, SPF/DKIM/DMARC, Mail path reconstruction, Timestamp analysis | Key Modules: Header Forensics, Mail Path, Infrastructure Risk

Day 8 Execution: Engineered DOM parser in `content.js` to extract raw email headers, sender display names, and recipient addresses from Gmail UI.
- Lahari.K — IOC & Infrastructure Lead

Domain Ownership: URL, Domain, IP extraction, Geolocation, ASN/ISP mapping, Malicious indicator detection | Key Modules: IOC Extraction, Geolocation, URL/Domain/IP Risk

Day 8 Execution: Built real-time URL link scanner inside `content.js`: inspecting all hyperlink anchor tags in email bodies for deceptive display-vs-href mismatches.
- Hemanth.P — Graph & Campaign Intelligence Lead

Domain Ownership: Attack Relationship Graph, Similarity detection, Campaign grouping & correlation | Key Modules: Graph Engine, Campaign Detection, Similar Attack Detection

Day 8 Execution: Designed lightweight payload compressor for transfer between Extension Service Worker and Frontend Cockpit via URL hash state.
- Abbas.N — Risk, Case & Response Lead

Domain Ownership: Dynamic Threat Score engine, Automatic Forensic Case creation, Recommended Response actions | Key Modules: Risk Scoring, Case Management, Response Recommendations

Day 8 Execution: Created extension message router handling `ANALYZE\_CURRENT\_TAB`, `REPORT\_CYBERCRIME`, and `GENERATE\_CANARY` actions.
- Moulaanbee.N — Frontend + Assistant + Simulator Lead

Domain Ownership: Dashboard, Visualizations (Graph, Map, Timeline), AI Chatbot, Email Threat Simulator, Overall UI/UX | Key Modules: Dashboard, AI Cybersecurity Assistant, Threat Simulator, Frontend Integration

Day 8 Execution: Designed and coded the Extension Popup interface (`popup.html` / `popup.js`) and in-Gmail floating forensic score shield overlay.

System Integration & Quality Verification

Cross-Module Integration: Successfully loaded unpacked extension in Chrome & Edge; verified automatic threat scoring badge injection inside Gmail.

Quality Assurance & Verification: All technical modules, automated unit test suites, and API contracts for Day 8 executed with 100% clean passes. Code reviewed and merged into main repository.

Prepared By: Threat Trace AI Development Team | **Project Lead:** Muniswami.Y | **Status:** SPRINT DAY 8 MILESTONE SIGNED OFF